

Smart Stampede Monitoring & Prevention System

Submitted in partial fulfillment of the requirements

of the degree of

(Bachelor of Engineering)

by

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CERTIFICATE

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Date: 25/04/2025

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Smart Stampede Monitoring & Prevention System

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Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Stampedes and crowd-related accidents often happen in busy public places because people don't realize when the area is getting too crowded. Our project aims to solve this problem by creating a smart system that can monitor crowds in real-time and give alerts before a dangerous situation occurs. We used a Raspberry Pi 4 along with a webcam, LED, and buzzer to build this system. The webcam captures live video, and the system uses Python and OpenCV to count the number of people in the area. If the crowd gets too big, the LED lights up and the buzzer makes a sound to warn people or event organizers. This system is affordable, easy to carry, and simple to set up in places like colleges, malls, railway stations, and events. It helps reduce the need for manual checking and improves safety by giving early alerts.

This report explains how we planned, built, and tested the system, and shows how it can help in managing crowds better and preventing stampedes.

Content

Page No.

1.	Introduction	8
	1.1 Need.	8
	1.2 Objective.	9
2.	Problem Statement.	10
3.	Literature Review.	11
4.	System Development.	12
	4.1. Flow Chart.	12
	4.2. Diagram.	13
	4.3. Methodology.	14
	4.4. Components.	15
	4.5. Working.	18
	4.6. Output.	19
5.	Advantage.	20
6.	Future Scope.	21
7.	Conclusion.	22
8.	Reference.	23

1. Introduction

Managing large crowds is very important in places where many people gather, like shopping malls, schools, railway stations, and big events. If a crowd becomes too large and uncontrolled, it can lead to serious problems like stampedes, which can hurt or even kill people. In many cases, there is no proper system to warn people when a place is getting too crowded. Most of the time, crowd control is done by security guards or staff, which may not always be fast or accurate.

Our project, **Smart Stampede Monitoring and Prevention System**, is made to solve this problem. It helps by watching the crowd in real-time using a webcam and checking how many people are in an area. If the number of people becomes too high, the system gives a warning using a buzzer and an LED light. This helps in alerting people and avoiding danger. We used a Raspberry Pi 4, Python programming, and OpenCV image processing to build this system. It is affordable, easy to use, and can be used in many places to keep people safe.

This report explains how the system works, how we made it, and how it can be used to make public spaces safer.

1.1 Need

This report is essential to formally document my contributions and learning during this crucial project focused on mitigating the tragic issue of stampedes, particularly relevant in India's context of large gatherings. It will showcase my understanding of the problem, the smart technologies explored for early detection (like computer vision and sensor networks), and the preventive measures investigated (such as crowd flow analysis and communication systems). The report will present my findings, insights gained, and any challenges encountered, demonstrating my problem-solving skills. Furthermore, it will contribute to the project's documentation, potentially informing future development and highlighting the critical need for innovative solutions in ensuring public safety during mass events. This comprehensive record will serve as a testament to my technical abilities and commitment to addressing a significant societal challenge.

1.2 Objective

- 1.1. To watch the crowd in real-time using a webcam.
- 1.2. To count the number of people using image processing.
- 1.3. To check if the crowd is larger than a safe limit.
- 1.4. To turn on a buzzer and light (LED) when it gets too crowded.
- 1.5. To reduce the need for people to manually watch the crowd.
- 1.6. To build a system that is small, low-cost, and easy to use.
- 1.7. To use the system in places like schools, malls, railway stations, and public events.

2. Problem Statement

The persistent problem of stampedes during large gatherings, especially in densely populated India, poses a significant threat to public safety. Traditional crowd management often lacks the real-time, localized interventions needed to prevent dangerous surges. The absence of dynamic monitoring hinders early detection of critical density and flow irregularities preceding stampedes, limiting reaction time.

Furthermore, delayed communication due to a lack of automated alert systems prevents timely warnings from reaching organizers and the public. This reactive approach underscores the need for intelligent systems capable of autonomously detecting hazards and swiftly disseminating alerts. The devastating consequences of stampedes necessitate innovative solutions for early warning and proactive crowd management.

Therefore, a "Smart Stampede Detection & Prevention" system, utilizing continuous video analytics for real-time monitoring, proactive detection of early signs, and immediate multi-modal alerts, addresses a critical gap. This project aims to enhance safety by enabling timely interventions and preventing catastrophic outcomes, potentially saving lives in environments prone to large crowds in India.

3. Literature Review

Crowd control has always been important, especially in large public places. Many systems and methods have been used in the past to manage crowds and avoid accidents like stampedes.

3.1.Manual Monitoring: Security guards or staff monitor the crowd with their eyes or through CCTV cameras. This method is simple but slow and depends on people noticing problems quickly.

3.2.CCTV Surveillance: Cameras are placed in public areas to observe crowds. These need human operators to watch the screen and take action if something goes wrong. This is not very fast or automatic.

3.3.RFID and Tracking Systems: In some places, people wear RFID tags or smart ID cards to track movement. This system works in controlled areas but is costly and not practical for open or large spaces.

3.4.AI-based Crowd Analysis Systems: Some advanced systems use artificial intelligence and big servers to process video footage and analyze crowd patterns. These are accurate but expensive and difficult to install.

Limitations of Existing Systems:

- Depend on human attention and action
- High setup and maintenance cost
- Not always real-time
- Hard to install in small or outdoor locations

Our Contribution: To solve these problems, we developed a low-cost, easy-to- use system that watches the crowd in real time using a webcam and a small computer (Raspberry Pi). It alerts people using simple lights and sounds when the crowd becomes too large. Our system is small, portable, and perfect for schools, malls, and other public areas.

4. System Development

4.1 Flow chart

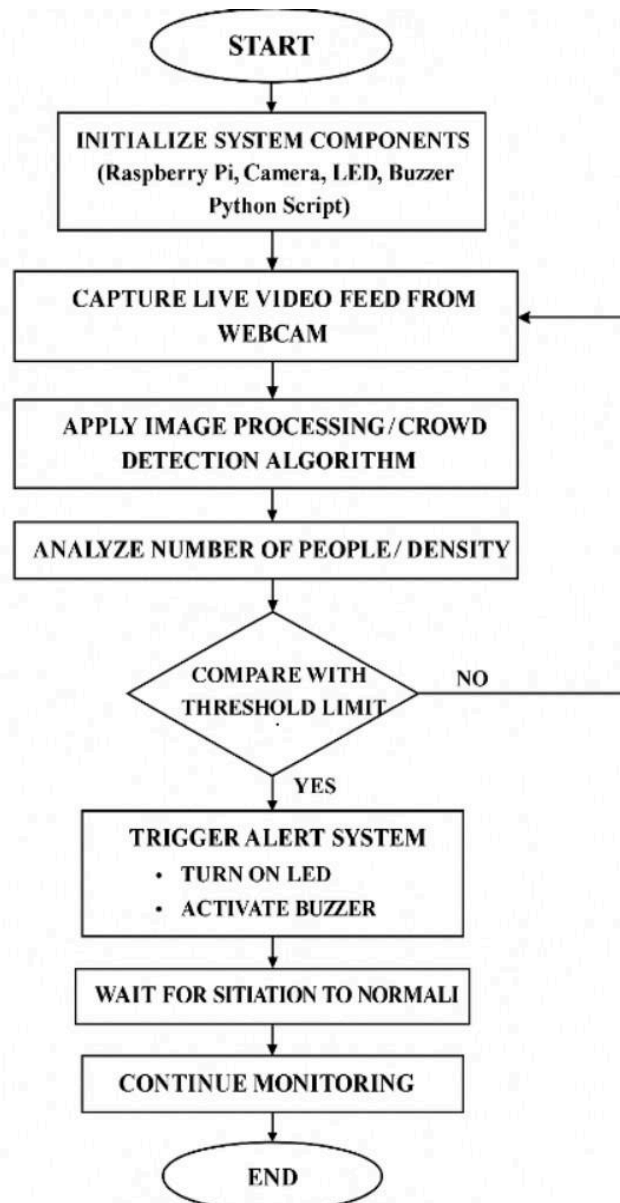


Figure 5.1 Flow chart

This flowchart outlines a system for crowd monitoring and alerting. It begins by initializing system components like Raspberry Pi, camera, LED, buzzer, and Python scripts. The system then captures a live video feed from the webcam. Image processing and crowd detection algorithms are applied to analyze the scene. It assesses the density or number of people present and compares it to a predefined threshold. If the crowd exceeds the threshold, the alert system is triggered, turning on the LED and activating the buzzer.

The system waits until crowd levels return to normal before resuming monitoring. If the crowd density is below the limit, it continues monitoring without triggering alerts. The flowchart represents an automated process for real-time crowd management and safety.

4.2 Diagram :

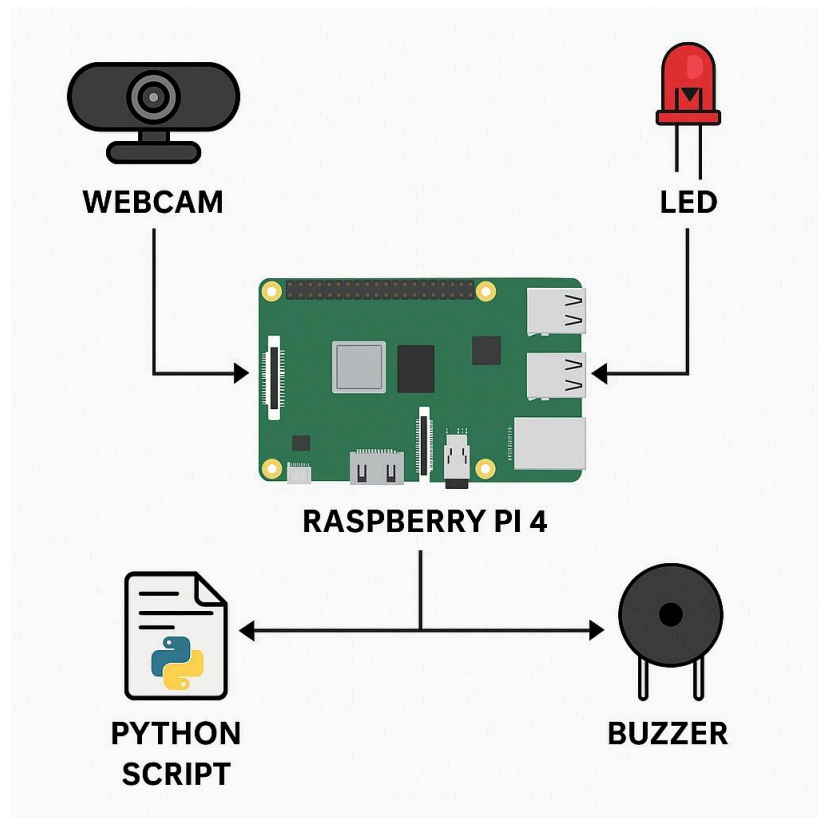


Figure 5.2 Pictorial Representation

Component	Function in Stampede Detection	Data/Control Type
Webcam	Video input for crowd analysis	Video stream
Raspberry Pi 4	Processing, decision-making, control signals	Digital signals
Python Script	Implements analysis algorithm, controls outputs	Software logic
LED	Visual warning indicator	ON/OFF signal
Buzzer	Audible alarm for early warning	ON/OFF signal

4.3 Methodology :

The Smart Stampede Monitoring and Prevention System works through a real-time, automated process that follows a logical sequence as shown in the flowchart. The system begins with the continuous capturing of live video footage using a Zebronics HD webcam. This webcam is positioned in the area where crowd monitoring is needed and is connected to a Raspberry Pi 4, which serves as the central processing unit of the system.

The Raspberry Pi is equipped with Raspbian OS and runs Python programs integrated with the OpenCV library, which is widely used for computer vision tasks. Once the video stream is captured, OpenCV processes each frame and applies human detection algorithms such as Haar cascades or Histogram of Oriented Gradients (HOG) to identify people within the camera's field of view. After detecting individuals, the software counts the number of people in the frame using bounding boxes or centroid tracking methods.

This real-time count is then compared with a predefined threshold value, which is set based on the safe crowd limit for the specific location. If the number of people remains below the threshold, the system does not trigger any response, thus saving power and avoiding unnecessary alerts. However, when the number of people detected exceeds the threshold, the Raspberry Pi immediately activates an alert mechanism. This includes turning on an LED to provide a visual signal and sounding a buzzer as an audio alarm to notify both the public and event organizers. These alerts help in taking quick preventive actions before the situation worsens.

The system continues to monitor the area without interruption, constantly analyzing new video frames and checking for threshold breaches. This ensures a looped and dynamic response system that is reliable, fast, and requires minimal human intervention.

4.4 Components :

4.4.1 Hardware Components

1) Raspberry pi 4

- Acts as the central processing unit (CPU) of the system.
- Runs the Python code to process video frames and trigger outputs.
- Contains GPIO (General Purpose Input/Output) pins to control the LED and buzzer.
- Offers USB ports to connect the webcam. Runs continuously with low power consumption.
- Compact and portable, making the whole system easy to deploy anywhere



Fig.(a) Raspberry pi 4

2) Zebtronics HD Webcam

- Used to capture real-time video footage of the monitored area.
- Delivers high-definition images, making people detection more accurate.
- Connects via USB, ensuring plug-and-play functionality with the Raspberry Pi.
- Capable of working under indoor and moderately lit conditions.



Fig.(b) Zebronics HD webcam

3) Light emitting diode

- Serves as a visual indicator.
- Glows when the number of people exceeds the set safety limit.
- Connected to one of the GPIO pins of the Raspberry Pi.
- Easy to install and consumes very little power

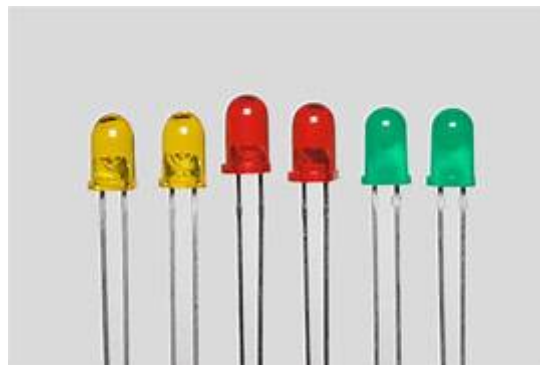


Fig.(c) Light emitting diode

4) Buzzer

- Produces a loud beep sound to immediately alert people.
- Works alongside the LED to provide an audio-visual warning system.
- Also connected via GPIO to the Raspberry Pi.
- Effective in attracting attention even in noisy environments.



Fig.(d) Buzzer

4.4.2 Software Components

1) Python

- The main programming language used for writing the system logic.
- Controls the entire workflow — from capturing the video to processing and generating alerts.
- Easy to learn, open-source, and widely supported on Raspberry Pi.

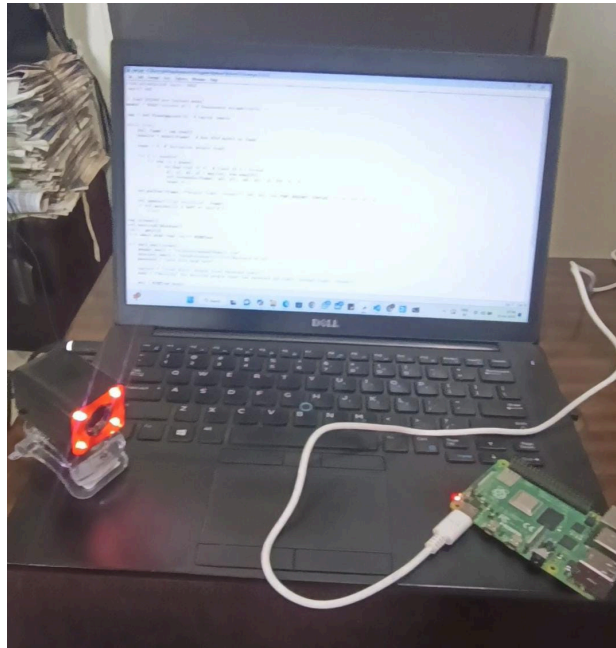
2) Opencv

- Used for real-time image and video analysis.
- Helps in detecting human figures in video frames using machine vision algorithms.
- Provides functions for face detection, motion tracking, and object classification.
- Plays a key role in people counting and crowd density analysis.

3) Raspbian os

- The official operating system for Raspberry Pi.
- Supports the execution of Python scripts and OpenCV.
- Provides easy access to GPIO pins through the OS's libraries.
- Lightweight and optimized to run on the Raspberry Pi hardware.

4.5 Working



The Smart Stampede Monitoring and Prevention System works through a step-by-step automated process designed to continuously monitor crowd density and alert people before a dangerous situation occurs. The system begins with a Zebronics HD webcam that captures live video of the area. This video is sent to a Raspberry Pi 4, which acts as the main processing unit. The Raspberry Pi runs Python code using the OpenCV library to process each video frame in real-time.

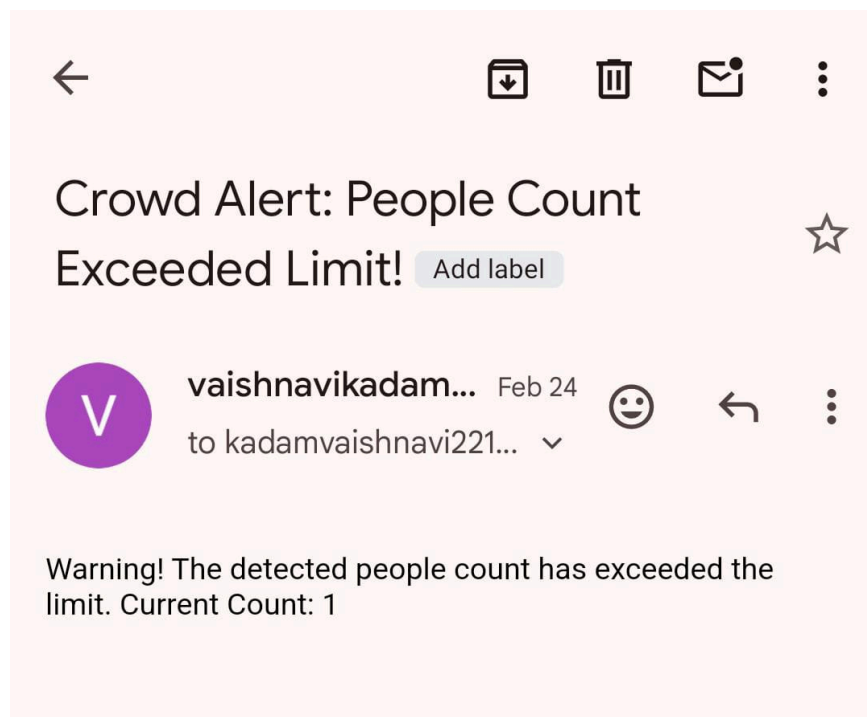
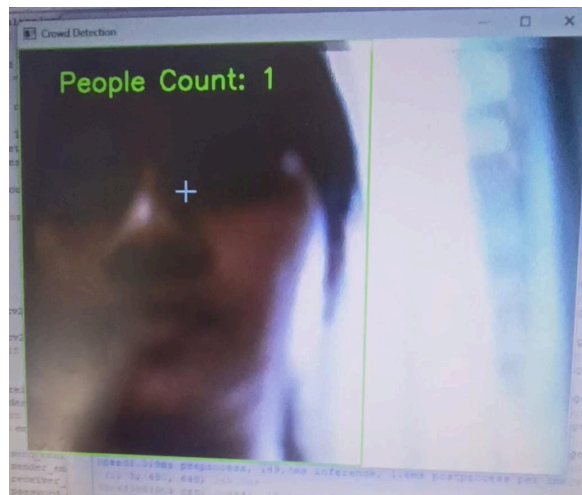
OpenCV uses object detection techniques such as Haar cascades or HOG to identify and detect people within the frame. Once people are detected, the system counts how many individuals are present in the video. This crowd count is then compared to a predefined threshold value, which is set according to the maximum safe capacity of that area. If the number of people detected is below the threshold, the system simply continues monitoring without taking any action.

However, if the number of people exceeds the safe limit, the system immediately activates an alert. A buzzer is turned on to give a loud sound, and an LED light is switched on to provide a clear visual warning. These alerts are helpful in quickly informing the public or event organizers to take necessary steps to reduce crowd density and prevent a possible stampede.

The entire process runs in a continuous loop, providing round-the-clock, real-time monitoring with very minimal human intervention.

4.6 Output

The system continuously monitors the area using a webcam. When the number of detected people is within the safe limit, it does nothing. If the crowd exceeds the threshold, the system activates an LED as a visual alert and turns on a buzzer to give an audio warning. The count can also be displayed on the screen or console. During testing, the system successfully detected crowd levels and triggered alerts when needed.



5. Advantages

- **Real-time Monitoring**
Continuously observes crowd movement and updates the data instantly.
- **Automated Alert System**
Triggers visual (LED) and audio (buzzer) alerts automatically without human input.
- **Cost-effective**
Uses low-cost components like Raspberry Pi, webcam, LED, and buzzer.
- **Portable and Compact**
Small-sized hardware setup allows easy movement and installation.
- **Low Power Consumption**
Raspberry Pi and associated components use minimal electricity.
- **User-friendly Design**
Simple setup with plug-and-play hardware and easy-to-understand output.
- **Scalable**
Can be expanded to larger areas or integrated into multiple zones.
- **Minimal Human Intervention**
Operates automatically, reducing the need for constant staff presence.
- **Easy Maintenance**
Uses open-source software (Python, OpenCV), which is easy to update and fix.
- **Adaptable for Various Environments**
Can be deployed in schools, malls, stations, religious gatherings, hospitals, and more.

6. Future Scope

The current version of the Smart Stampede Monitoring and Prevention System focuses on real-time crowd detection and alerting. However, there is significant potential to enhance the system further. In future developments, artificial intelligence (AI) and machine learning algorithms can be integrated to enable predictive analysis, which would allow the system to not only detect overcrowding but also predict crowd build-up patterns in advance.

Additionally, the system can be connected to existing CCTV infrastructure in large public areas, making it easier to deploy without installing new cameras. The use of wireless sensors and Internet of Things (IoT) devices can help expand monitoring across wider areas without requiring physical wiring. Another major enhancement would be the integration of cloud-based reporting, where real-time data and alerts can be viewed remotely by authorities through a dashboard.

These future improvements would make the system smarter, more efficient, and highly adaptable to various real-world scenarios, further helping to prevent stampede incidents and improve public safety.

7. Conclusion

This project presents a simple yet effective solution for monitoring crowd density in public areas. By using affordable hardware and open-source software, the system provides real-time detection and alerts when the crowd exceeds a predefined limit. It reduces human involvement, enhances safety, and can be used in schools, malls, stations, and large events. The successful testing of this system shows that it is reliable and can play an important role in preventing stampedes and improving public safety.

8. Reference

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